

Khulna University of Engineering & Technology
B.Sc. Engineering 4th Year 1st Term Examination, 2022
Department of Materials Science and Engineering

MSE 4101
(Modern Iron and Steel Making)

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) A blast furnace pig iron of composition of Fe-93.0%, C-3.6%, Si-2%, Mn-0.5%, P-0.9%. The flux amounts to one-fourth the weight of the ore. Consumption of coke is 780 kg/ton of pig iron. The gases carry 2 parts CO to 1 part CO₂. Assume that 99.0% of the iron is reduced and 1% iron is slagged. Calculate (20)
(i) the weight of the ore required to produce one ton (1000kg) pig iron.
(ii) the weight of slag made from one ton pig iron and its composition
(iii) volume and composition of blast furnace gas.
(b) Discuss the advantages that have been made in modern blast furnaces to reduce coke consumption and to increase their productivity. (15)
2. (a) Compare and contrast HYL III and HYL IV process. Out of these two processes select the most suitable one for Bangladesh and justify your selection. (15)
(b) Mention the physico-chemical principles of ferro-silicon smelting. (12)
(c) Explain the fundamental of sintering process of iron ores. (08)
3. (a) Explain the impact of CaO flux for the breakdown of silica with diagram. (10)
(b) Explain the weight type gas cleaning circuit with appropriate diagram. (10)
(c) Explain the electro chemical nature of slag metal reaction as metal is non polar and molten slag is ionic in character. (15)
4. (a) Describe the thermal, physical and chemical profiles in blast furnace with proper diagram. (15)
(b) How softening-melting test simulate the behavior of iron oxides in the cohesive zone? (10)
(c) Is sulphur reduction economically feasible in a blast furnace? Justify your answer. (10)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

5. (a) How are sulphur and phosphorous removed during the refining period of steel melting? (15)
- (b) Why is it important to have short areas during the Bore-in phase of the melting process in steel scrap? Explain in brief. (10)
- (c) Considering a scrap carbon content of 0.25%, silicon content of 0.25% and manganese content of 0.70%, along with the use of 20 kg of net coke per ton of scrap (with an 80% yield), what is the theoretical amount of Nm^3 oxygen required to achieve a desired carbon content of 0.08% when tapping 150 tons of scrap? (10)

6. (a) Calculate the total interfacial area of metal droplets in the slag-metal emulsion at steady state decarburization period of a BOF blow. Do the calculation for 1 ton of liquid steel in the bath. (10)

Assume: (i) The emulsion contains 7% of bath metal as droplets.

(ii) The droplets of metal are spherical.

(iii) The size distribution of the droplets are as follows:

Diameter (d), mm	0.2	1	2	5	10
Fraction of total number (N) of droplets	0.05	0.2	0.5	0.2	0.05

- (b) Explain why and how pre-treatment of B/F liquid iron is performed before charging into the BOF for making steel. (15)
- (c) "Look after your slag, the steel will look after itself"-explain your reason for this statement. (10)
7. (a) One steel plant is using an EAF to produce carbon steel from scrap. Now, the management wants to produce stainless steel and is seeking your advice. Which route will you suggest? Give reasoning to justify your answer with detail description of the suggested method. (18)
- (b) What is the difference between deoxidation and degassing? Among the existing degassing processes, which one you will select for a steel plant producing 100 tons of liquid steel in each heat? Justify your answer. (17)
8. (a) Why is continuous casting gradually replacing the conventional ingot casting process? Describe the metallurgical aspects of continuous castings of steel. (15)
- (b) What will you do or suggest if the following defects arises in billet- (10)
- (i) Surface crack (ii) Rhomboidity, (iii) Inclusions and (iv) Pinholes.
- (c) What is meant by clean steel? Synthetic slag practice is principally employed to obtained clean steels and to desulphurise molten steel-explain in brief. (10)

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B.Sc. Engineering 4th Year 1st Term Examination, 2022
Department of Materials Science and Engineering

MSE 4103
(Engineering Alloys and Materials Selection)

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
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(iii) Assume reasonable data if any missing.

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) Illustrate the carbide phases present in stainless steel in the phase diagram. Explain the role (08+07)
of these phases regarding the mechanical properties of stainless steel.
(b) Describe the heat treatment procedures of the 17-7PH grade precipitation hardened stainless (12)
steel with a suitable diagram.
(c) High-strength low alloy steel is termed high-purity steel. Explain how this alloy is an ideal (08)
choice for making automobile bodies.
2. (a) Explain the importance of selecting suitable heat treatment temperature and quenching rate (06+06)
regarding tool steels.
(b) “For the highest requirements on dimensional stability, sub-zero treatment in liquid nitrogen (12)
is recommended after quenching and after each tempering”-Justify the statement.
(c) Explain how maraging steel attains high tensile strength? (11)
3. (a) Between 7075 and 7130 aluminum alloys, which one will you choose for making automotive (10)
parts when you don't have access to solution hardening and you can't cold roll the product?
(b) Select a suitable material with appropriate reasoning(s) for spring applications from the (20)
following Titanium-based materials:
(i) Grade 2, (ii) Grade II, (iii) Ti-3Al-2.5V, (iv) Ti-6Al-4V and (v) Ti-3Al-8V-6Cr-4Mn-4Zr.
The selected materials must fulfill the following criterions:
* Must have good forming and shaping characteristics
* Must have good resistance to hydrogen embrittlement
* Must have good ductility
* Must be solution treatable.
(c) Between Muntz metal and Naval Brass, which one is suitable for salt-water environment (05)
applications and why?
4. (a) For manufacturing a turbine blade in jet engine, what types of alloys are most appropriate? (15)
Write down the common characteristics and explain with neat sketches of microstructure how
creep resistance is obtained.
(b) Validate the underlying mechanism for shape memory alloy with example. (10)
(c) What is hydrogen embrittlement? How does hydrogen embrittlement affect the properties of (10)
Titanium alloys?

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

5. (a) "Iteration, looping back to explore alternatives, is as essential part of the design process" – (10)
Explain the assertion.
- (b) Derive material index for a light, stiff beam as shown in figure 5(b). The length L is specified (13)
but the cross-section area is not.
- (c) Discuss the materials selection process strategy using window glass as an example. (12)

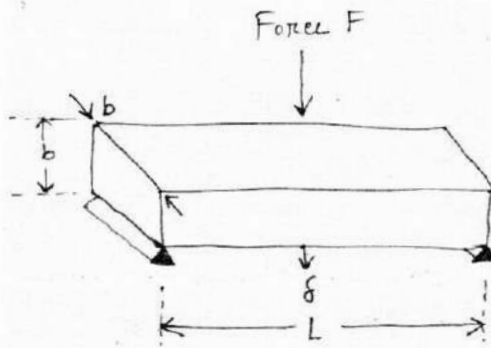


Figure 5(b)

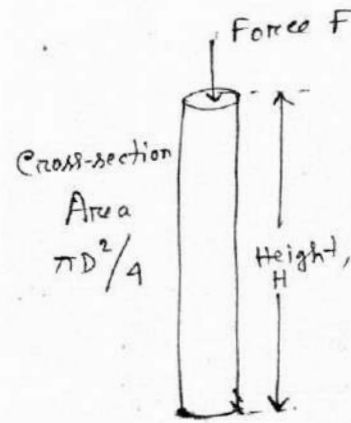


Figure 6(c)

6. (a) Explain the concept of penalty function. (10)
- (b) Derive the expression for the gain in bending stiffness when a solid square beam of side b_0 is (10)
expanded at constant mass to a larger square section with a prismatic cell structure.
- (c) It is sometimes suggested that architects live in the past; that in the 21st century they should (15)
be building with fiberglass, aluminum alloys and stainless steel. Consider the possibility of
replacing wooden or concrete column show in figure 6(c) of a building with AA 6061
aluminum alloys or AISI 304 stainless steel. The table below summarizes the needs.

Function	Column
Constraints	Must not fail by compressive crushing
	Must not buckle
	Height H and compressive load F specified
Objective	Minimize material cost C
Free variables	Diameter D
	Choice of Material

Data for the four candidates for the column are listed below. Use these to identify the best candidate material when $F=10^5\text{N}$ and $H=3\text{m}$.

Material	Density, $\rho(\text{kg/m}^3)$	Unit-cost, $C_m(\$/\text{kg})$	Modulus, $M(\text{MPa})$	Compressive strength, $\Gamma_c(\text{MPa})$
Wood	700	0.50	10,000	25
Poured Concrete	2300	0.08	20,000	13
AA 6061-T6 Al-alloy	2700	4.45	69,000	155
AISI 304 stainless steel	7850	1.00	200,000	250

7. (a) Explain compressive deformation behavior of foams with neat sketches. (10)
- (b) Among single charts, 4-quadrant charts and indices including SF charts, which one do you think is better? Justify your answer. (10)
- (c) Derive the expression for the shape-efficiency factor Φ_B^e for stiffness-limited design for a beam loaded in bending with each of the three sections listed below for the figure 7(c). Do not assume that the thin-walled approximation is valid. (15)
- (i) A closed circular tube of outer radius $5t$ and wall thickness t ,
- (ii) A channel section of thickness t , overall flange width $5t$ and overall depth $10t$, bent about its major axis; and
- (iii) A box section of wall thickness t and height and width $h_1=10t$.

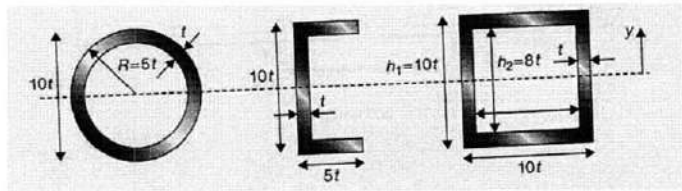


Figure 7(c)

8. (a) Explain the concept of percolation and using suitable example, examine its importance as a design tool for hybrid materials. (10)
- (b) How material, function, shape and process are interrelated? Explain with suitable examples. (10)
- (c) Using the figure 8(c) devise hybrids to fill the holes, with the following applications in mind: (15)
- (i) Sheet steel (as used in car body panel, for instance) is prone to lightly damped flexural vibrations. Devise a hybrid sheet that combines the high stiffness, E , of steel with high loss coefficient, n .
- (ii) Devise a hybrid with low modulus E and low loss coefficient, n .

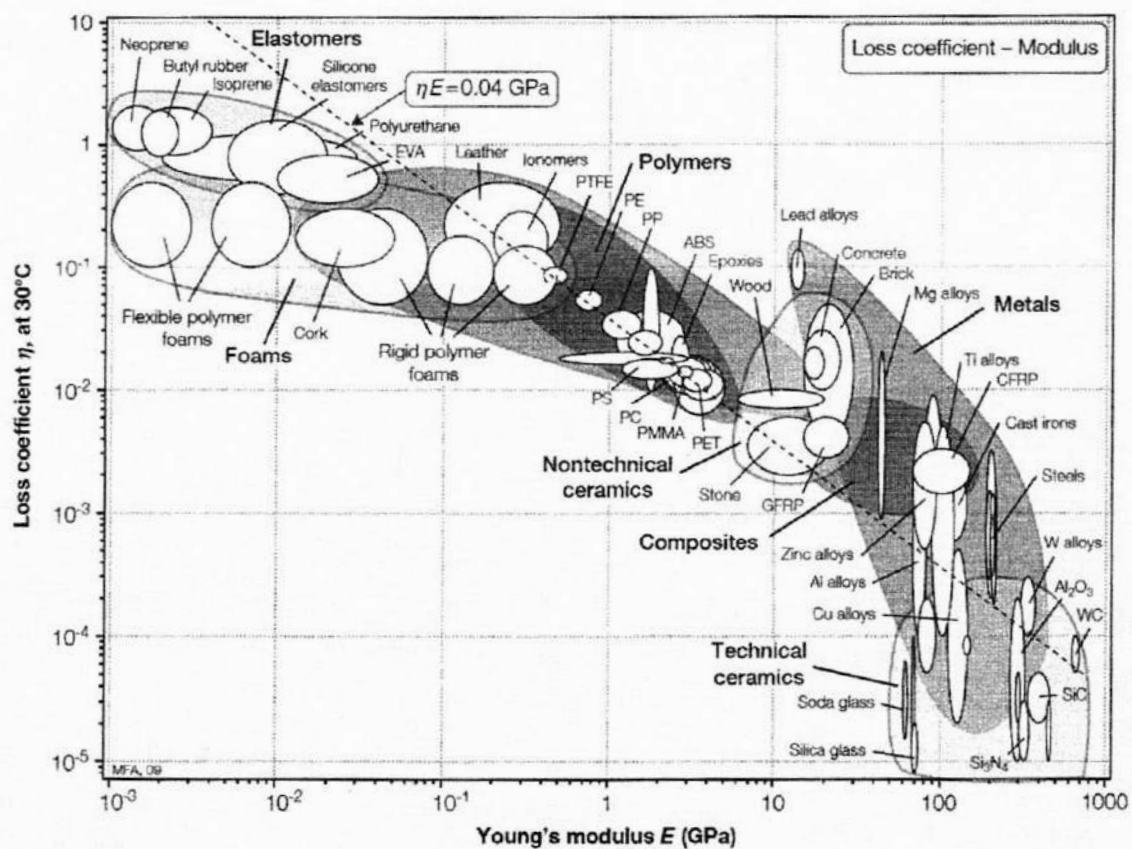


Figure 8(c)

Khulna University of Engineering & Technology
B.Sc. Engineering 4th Year 1st Term Examination, 2022
Department of Materials Science and Engineering

MSE 4105
(Polymer and Composite Materials)

Time: 3 hours

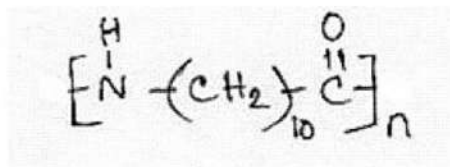
Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) Design and explain a fabrication process that is used to form polymeric bag. (15)
(b) Explain why glass transition temperature is important during in service life of polymer. (12)
(c) Explain the usages of stabilizers as polymeric additive. (08)
2. (a) The thermoplastic elastomers have replaced conventional thermoset elastomers for the electrical insulation of automotive components. Justify the statement with diagram. (15)
(b) Localized strengthening is important during tensile loading of a polymer. Explain in brief. (10)
(c) Adhesive bonding offers some advantages over joining technology like welding. Justify the statement. (10)
3. (a) Explain the factors that influence the mechanical properties of semi-crystalline polymers. (10)
(b) Appreciable tensile deformation of semi-crystalline polymer produces a highly oriented structure. Justify the statement with diagram. (15)
(c) What are the polymer additives and why these additives are used in polymer processing? (10)
4. (a) If the number average degree of polymerization, X_n for nylon is 100 and $M_w = 18300$. What is its polydispersity? Draw the molecular weight curve for this polymer. (13)



- (b) Describe the stress-strain behavior of different polymeric materials with proper diagrams. (13)
- (c) Differentiate between linear polymer and crosslink polymer. (09)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

5. (a) What are the functions of matrix and reinforcements in composite? (08)
- (b) Explain the modulus of composite in transverse direction with equation. (10)
- (c) Explain the factors affecting the properties of composites. (09)
- (d) Write down and describe the defects commonly found in composites. (08)
6. (a) What are the basic characteristics of carbon and graphite fibers? (09)
- (b) Define whiskers. Why whiskers are very expensive? (10)
- (c) Compare between thermoset and thermoplastic matrix with examples. (06)
- (d) Describe the pultrusion process of polymer matrix composite fabrication with figure. (10)
7. (a) What is liquid state fabrication of MMC? Explain different types of infiltration process of liquid state MMC fabrication. (10)
- (b) Write down the advantages and disadvantages of in-situ fabrication of MMC. (08)
- (c) How can composite materials bear higher load than conventional materials. Explain with necessary figures. (10)
- (d) A thermoplastic matrix contains 35% weight of glass fiber. If the density of matrix is 1.2 gm/cm^3 and density of glass fiber is 2.6 gm/cm^3 , then calculate the density of composite. (07)
8. (a) Why is natural fiber polymer composite more ecofriendly than synthetic fiber polymer composite? (12)
- (b) Write short notes on: (06)
- (i) interfacial shear strength test and
- (ii) creep of composites.
- (c) Describe the fracture toughening mechanisms in composite. (07)
- (d) Mention the 'big five' methods of non-destructive testing of composite. Among them select a suitable one to detect volume defects. Justify your answer. (10)

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MSE 4107
(Computational Materials Science)

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) Discuss the Hohenberg-Kohn theorems with necessary figures. Proof that the external potential is (05+10)
a unique functional of the density.
- (b) Explain the role of electron density in the DFT. (10)
- (c) Explain the antisymmetry principle by choosing a suitable 3×3 matrix and suggest a suitable (10)
method to overcome this limitation in Hartree method.
2. (a) Write the mathematical expression of the Hamiltonian operator and explain it. (10)
- (b) List out the ways to solve the KS equation. Explain with illustration of a typical DFT procedure (02+13)
by self-consistent loop.
- (c) "The quality of a DFT run is critically determined by how closely the approximate exchange- (10)
correlation energy reproduces the exact value"- Justify the statement.
3. (a) Band gap underestimation is a major limitation of GGA functional. Discuss why this limitation (06+06)
arises in DFT calculation and suggest suitable remedies for this problem.
- (b) Identify the exchange-correlation functionals X and Y from Table 3(b) with proper reasoning. (10)

Table 3(b)

Parameter	Experimental Data	Functional	
		X	Y
a(Å)	3.244	3.189	3.326
C(Å)	5.199	5.237	5.418

- (c) For a cellular automaton, the data is represented as below: (02+11)

000 001 010 011 100 101 110 111

The above data corresponds to the generation of a specific "Rule". Identify the "Rule" and show how this "Rule" can be achieved.

4. (a) Explain Hesselbarth and Gobel's model of cellular automata for the recrystallization of materials. (13)
- (b) Figure 4(b) shows the convergence test for cutoff energy. Select the proper cutoff energy from the (08)
figure and explain your selection.

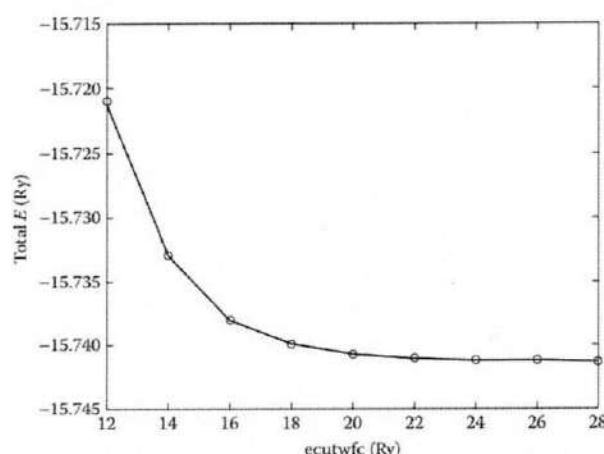


Figure: 4(b)

- (c) The Figure 4(c) represents consecutive snapshots from a 100×100 Game of Life. Referring to Figure 4(c), explain how these snapshots arise from Game of Life. (10)

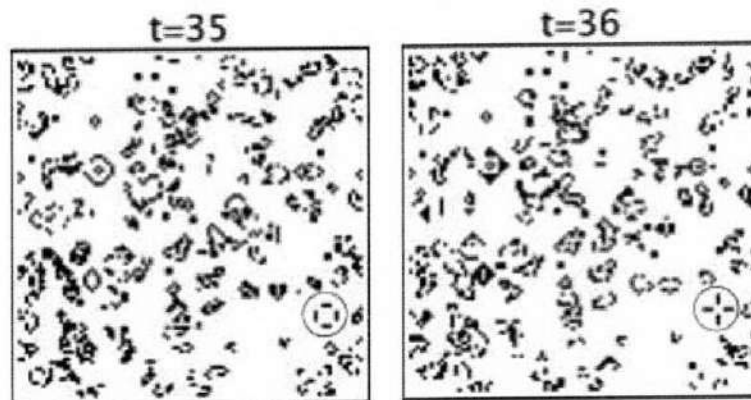


Figure: 4(c)

- (d) List out the applications of cellular automata. (04)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

5. (a) What is the role of potential functions in a Molecular Dynamics simulation? Explain several features of Lennard-Jones potential with a suitable diagram. (05+10)
- (b) "Difference of material properties originate from different atomic interaction". Discuss the statement. (11)
- (c) Write down the general expression of Mie potential and prove that minimum energy is located at interatomic distance, $r_m = (m/n)^{1/(m-n)} \sigma$, where the parameters have usual meanings. (09)
6. (a) Choose a suitable potential for metals and describe the potential with schematic diagram. (12)
- (b) Explain the basic principle of the predictor-corrector algorithm. Also, mention the pros and cons of it. (10+05)
- (c) What is potential Cutoff? Explain why potential Cutoff is necessary. (08)
7. (a) Explain the velocity rescaling process in the NVE ensemble with an appropriate example. (14)
- (b) Explain how structural properties are calculated from the Molecular Dynamics. (14)
- (c) Discuss how Molecular Dynamics find minimum energy using simulated annealing. (07)
8. (a) How does the Ising Model determine magnetization in materials using the Metropolis Monte Carlo method? (15)
- (b) Describe the principle of Monte Carlo method. Also, explain its limitations. (12)
- (c) Ensembles are of different types. Choose a suitable ensemble for Monte Carlo and discuss the ensemble. (08)

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MSE 4109

(Electronic, Magnetic and Optical Properties of Materials)

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) Select a copper alloy for the windings of an electromagnet designed to create very high magnetic field. The windings must sustain stresses much higher than pure Cu can handle. Explain what kind of alloying elements you will use to increase in strength with minimum decrease in conductivity. (10)
- (b) According to Debye's model 'heating a cup of tea in a microwave oven is due to orientation polarization'-explain in brief. (15)
- (c) A composite material is made of alternating thin lamellae of copper (resistivity $1.65 \times 10^{-6} \text{ ohm-cm}$) and a niobium-titanium alloy (resistivity $7 \times 10^{-5} \text{ ohm-cm}$) of equal thickness. What is the resistivity of this composite (i) parallel to the lamellae & (ii) perpendicular to the lamellae, if current is passed through the composite in this direction respectively? (10)
2. (a) Derive the expression for conductivity of metals. Define drift velocity, mobility and relaxation time for free electron. (12)
- (b) Describe the microscopic mechanisms of polarization in dielectrics. (15)
- (c) The dielectric constant of LiF is 9.27 at low frequencies and approaches 1.9 at high frequencies. What are the percentage contributions of electronic and ionic polarization to the low frequency polarization? Which ion contributes most to the electronic polarization? (08)
3. (a) Outline the various drawbacks of classical free electron theory of metals. (08)
- (b) Prove the Heisenberg uncertainty principle applying the Schrodinger's wave equation, consider the motion of an electron in free space. (12)
- (c) Calculate the probability of an electron tunneling through a potential barriers. Consider an electron with an energy of 2eV impinging on a potential barrier with $V_0=20\text{eV}$ and a width of $3\text{\AA}$. (08)
- (d) Show that the maximum in the radial probability function for the 1s electron state in hydrogen occurs at $r = a_0$. (07)
4. (a) What wavelength photons are required to excite electrons from the ground state to the first excited state in a three-dimensional energy well of dimensions $7\text{nm} \times 7\text{nm} \times 6\text{nm}$? In Joules, what are the energies of the two electron states? Is either of these states degenerate? (10)
- (b) For a free electron, calculate the density of quantum states per unit volume ($/\text{cm}^3$) over the energy range of (i) $0 \leq E \leq 2 \text{ eV}$ and (ii) $1 \leq E \leq 2 \text{ eV}$. (10)
- (c) What is the physical significance of wave function? Using Schrodinger equation, explain the wave-particle duality of electrons. (15)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

5. (a) Write down some important properties of magnetic field lines with necessary figures. (10)
- (b) Define magnetic susceptibility and relative permeability with examples. Hence, derive the relation between them. (04+08)
- (c) Explain the temperature dependency of different types of magnetism. Why is TiCl_2 paramagnetic but TiO_2 diamagnetic? (05+08)
6. (a) You have a composite in which the Nb-Ti filaments comprise 25% of the cross-section, the copper matrix 75%, and you use it to carry a current of 100 amps. If the resistivity of copper and Nb-Ti are 2×10^{-8} and 1×10^{-6} ohm-m respectively, at room temperature, and 4×10^{-11} and 8×10^{-7} at 9.1K, how much current will each phase carry at each of those temperatures? How much current will each carry at 9K when the Nb-Ti becomes superconducting? (15)
- (b) Describe the hysteresis behavior of soft and hard magnets. (10)
- (c) Define easy axis. Explain magnetic anisotropy for BCC and FCC single crystals for Fe and Ni respectively. (02+08)
7. (a) Explain reflection and refraction process with figures. (08)
- (b) Define absorption process. Describe the absorption of light by inter-band and intra-band transitions. (12)
- (c) A beam of red light with a wavelength in vacuum of 650nm enters an insulator with an effective dielectric constant ϵ_r of 1.7 at the frequencies of visible light. What will be the frequency, angular frequency, velocity and wavelength of this light? (10)
- (d) Describe the photoconductivity phenomenon. (05)
8. (a) Briefly explain Tyndall effect with an example. Differentiate between direct and indirect bandgap. (02+08)
- (b) "In the IR region, metals with large electrical conductivity are good reflectors"- prove the statement mathematically. (15)
- (c) Explain the working principle of LASER with necessary diagram. (10)

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MSE 4103
(Engineering Alloys and Materials Selection)

Time: 3 hours

Full Marks: 210

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Section B

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Explain the assertion.
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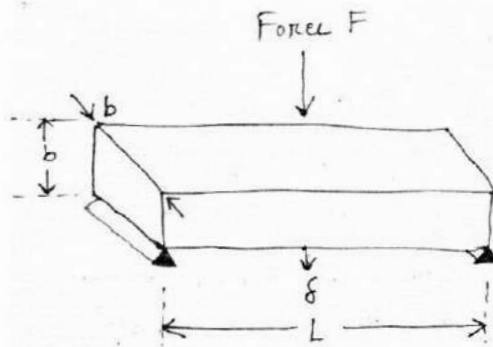


Figure 5(b)

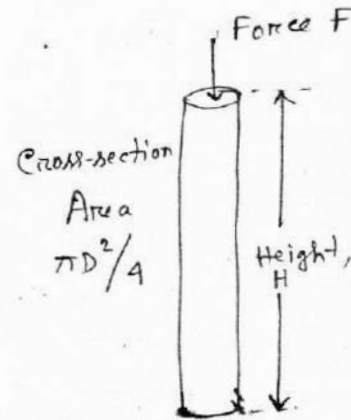


Figure 6(c)

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- (i) A closed circular tube of outer radius $5t$ and wall thickness t ,
- (ii) A channel section of thickness t , overall flange width $5t$ and overall depth $10t$, bent about its major axis; and
- (iii) A box section of wall thickness t and height and width $h_1=10t$.

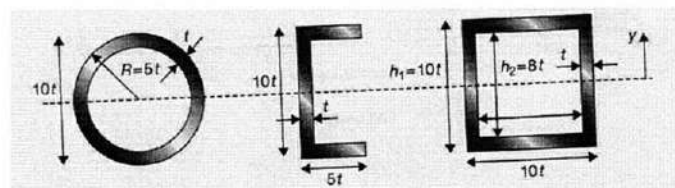


Figure 7(c)

8. (a) Explain the concept of percolation and using suitable example, examine its importance as a design tool for hybrid materials. (10)
- (b) How material, function, shape and process are interrelated? Explain with suitable examples. (10)
- (c) Using the figure 8(c) devise hybrids to fill the holes, with the following applications in mind: (15)
- (i) Sheet steel (as used in car body panel, for instance) is prone to lightly damped flexural vibrations. Devise a hybrid sheet that combines the high stiffness, E , of steel with high loss coefficient, n .
- (ii) Devise a hybrid with low modulus E and low loss coefficient, n .

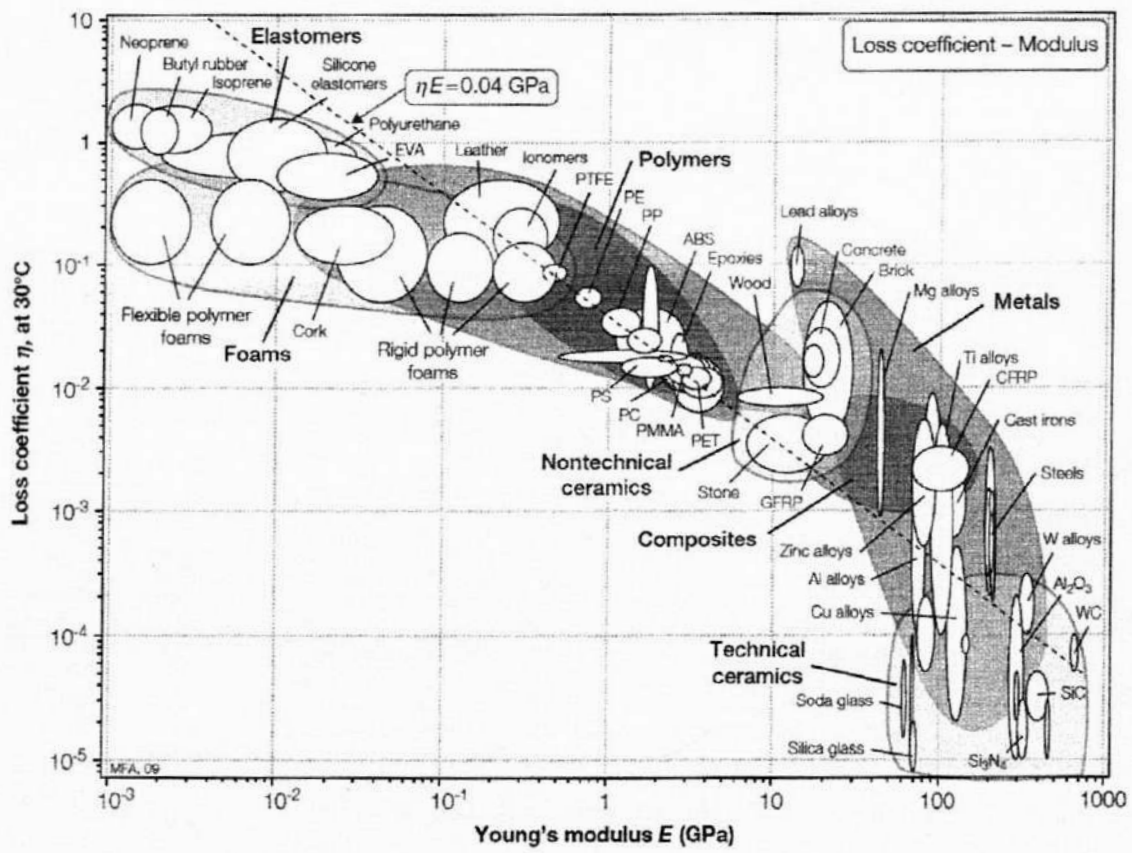


Figure 8(c)